

Cornea and ocular surface: a layered decision framework

Parts 1 and 2 | Think in layers and interacting barriers

A surface complaint can have tear, epithelial, neural and deeper causes.

Interacting protection systems

Lid closure and blink

Tear-film stability

Epithelial repair / limbus

Sensory and trophic support

Original map: protection depends on lids, tears, epithelium and neural support.

Start with symptom timing, blink, lid closure, medication exposure and the epithelial phenotype. Variable blur after blinking suggests an unstable optical surface, while conjunctivalisation suggests a different repair problem. Multiple mechanisms can coexist. Optimize the surface before irreversible refractive planning, because unstable tears and epithelium can distort both measurements and perceived surgical benefit.

Use imaging to answer a compartment question

A sharp cross-section does not provide an organism or a histological diagnosis.

Tool output	What it can answer
AS-OCT section	Depth, thinning and posterior-layer separation
Serial thickness profile	Structural progression or recovery
Hyperreflective infiltrate	Location, not organism identity
Limited penetration or shadow	Important uncertainty remains

Choose imaging from the suspected layer and recognize its limitations. A deep opacity can conceal structures behind it, and image contrast is not tissue culture. Combine structural information with slit-lamp findings and appropriate sampling. The most useful test is the one that changes the unresolved decision rather than merely producing a detailed picture.

Peripheral thinning has different biological drivers

An intact epithelium and lipid edge differ from active ulceration.

Pattern	Reasoning direction
Lid telangiectasia and meibomian dysfunction	Surface inflammation and evaporative disease
Slow peripheral thinning with lipid and intact epithelium	Consider Terrien degeneration
Active epithelial ulcer and rapid stromal loss	Investigate infection or immune destruction
Pain and adjacent scleral inflammation	Seek a systemic inflammatory driver

Assess time course and the relation between epithelial loss, vessels and thinning. Not every peripheral lesion belongs to an antibiotic pathway, and not every noninfectious lesion requires the same immunosuppression. A scar pattern is an anatomical endpoint; identify the process that is still active.

Tissue rescue and optical rehabilitation are staged goals

Restoring integrity does not guarantee a clear, regular optical surface.



Urgent tissue rescue can precede complete control.

Later optical planning revisits disease, geometry and visual potential.

Original clinical sequence: biological control and structural rescue precede stable optical planning.

Therapeutic or tectonic grafting can remove infected tissue or preserve globe integrity. Later rehabilitation depends on recipient disease, healing, astigmatism and posterior visual potential. An urgent graft can succeed structurally yet leave irregular optics because tissue rescue takes priority over refractive precision. Persistent active disease also increases the risk of later optical surgery.

A prosthesis requires a durable care system

Tertiary rescue changes the surveillance burden rather than removing it.

KPro consideration	Long-term consequence
Repeated transplant failure or unsuitable recipient	May justify tertiary prosthesis assessment
Surface and lid suitability	Determines which reconstruction is feasible
Infection, melt and membrane risk	Needs continuing specialist prevention and review
Glaucoma assessment difficulty	Pressure and nerve surveillance remain essential

Keratoprosthesis is an advanced referral-level option for selected severe corneal blindness. Discuss the eye, patient and follow-up system together. Optical restoration can be limited by glaucoma or retinal damage, and the device requires sustained care. This page describes selection principles, not an operative recipe.